

Mentor-mentee matching tool

Volodymyr Vyshnevetskyi, Pooja Vashisth

July 27, 2026

Abstract

Your abstract.

1 Introduction

Existing mentor–mentee matching systems typically optimize only one part of the problem at a time: either pairwise preference satisfaction via stable matching, approximate similarity via clustering, or high-level platform features such as communication and scheduling [1–3, 5, 7]. Two-sided matching models provide rigorous guarantees but assume one-to-one matches, rely on ordinal preference lists, and do not natively handle multi-mentee groups or rich profile features [3]. Clustering-based approaches can scale to large cohorts, yet they generally treat mentors as fixed centroids and mentees as points in a purely similarity-driven space, with limited support for hard constraints such as capacity, availability, or program-specific rules [5]. Broader work on co-clustering and constrained clustering demonstrates how side information can be incorporated into grouping, but it is rarely instantiated for mentor–mentee contexts with mixed numerical, categorical, and textual features alongside explicit program design requirements [6].

In contrast, this work treats mentor–mentee group formation as a joint optimization problem over three tightly coupled components: (i) mentor–group alignment on research interests, experience, and availability; (ii) intra-group structure among mentees; and (iii) program-level constraints on group size, capacity, and diversity. Mentee and mentor profiles are encoded as combined vector representations, while categorical and numerical attributes (e.g., education level, program type, experience level) are handled through appropriate distance metrics and constraint terms [4, 6]. The matching algorithm optimizes a single objective that balances (a) similarity to the mentor and the group’s research focus, (b) satisfaction of hard constraints such as mentor capacity, time-zone compatibility, and maximum group size, and (c) controlled diversity within each group. Diversity is explicitly parameterized through attributes such as academic level, field of study, institution or department, experience level, and soft skills.

At the system level, the proposed framework emphasizes an end-to-end workflow implemented over a normalized schema capturing user roles, rich mentor and mentee profiles, questionnaire responses, learned embeddings, and post-hoc feedback. This design enables full integration of data collection, matching, and evaluation within a single pipeline [2]. It further supports tracking of group performance over time and iterative refinement of the optimization objective based on observed satisfaction and outcome scores, rather than relying on static, hard-coded rules [1].

2 Related Work

A line of work models mentor–mentee assignment as a two-sided matching problem, where mentors and mentees provide ranked preference lists and the goal is to find a stable or near-optimal matching. [3] Haas and Hall compare variants of the Deferred Acceptance algorithm, approximation methods, and heuristic local search and genetic approaches under incomplete preferences and ties, benchmarking them against optimal solutions obtained via constraint optimisation on real mentor–mentee data. [3] These models operate on ordinal preferences, assume one-to-one relationships between mentors and mentees, and do not incorporate richer profile information such as research interests, background attributes, or questionnaire-based readiness scores, which limits their direct applicability to multi-mentee group formation where a mentor supervises a cohort rather than a single mentee.

Clustering-based systems and platform-oriented mentorship tools take a different approach. Board 47 presents a modified K-means algorithm in which mentors act as fixed centroids and mentees are assigned based on similarity in declared interests and experience, with the aim that at least 80% of students are matched to mentors sharing multiple interest areas and comparable experience. [5] The work demonstrates clustering-based matching, but it is evaluated on artificial data, uses an abstract similarity function without explicit vector modelling, and does not address group-internal diversity or capacity and scheduling constraints. [5] Other studies focus on building full mentorship platforms: Verma et al. describe an educational mentorship system that integrates a modern web frontend with collaborative filtering to connect mentees to subject-matter experts, [7] while “Mentor Connect” combines automated matching with scheduling, video communication, and analytics for career development. [1] A web-based peer mentoring platform for second-year education students integrates mentor matching with messaging, progress tracking, and study tools, and is evaluated using descriptive surveys of around one hundred participants. [2] Across these platforms, the matching logic is usually similarity-based and pairwise, and there is limited attention to explicit optimisation of group composition under constraints or to modelling research readiness and background diversity.

From the clustering literature, constrained and co-clustering approaches show how side information can be incorporated into grouping decisions. Sharma et al. survey coclustering and constraint clustering methods for text documents, including information-theoretic coclustering models that jointly cluster documents and word constraints using probabilistic formulations. [6] Okabe and Yamada propose a boosted constrained K-means algorithm that integrates background-knowledge constraints and metric learning; their method balances the simplicity of constrained K-means with ideas from boosting-based metric learning and demonstrates improved performance over standard constrained clustering techniques. [4] These methods illustrate how constraints and side information can guide clustering, but they are not tailored to mentorship settings and do not directly address mentor capacities, time-zone-aware availability, or questionnaire-based measures of research readiness and collaboration skills.

Beyond algorithms, prior work and practice discuss which human factors matter for mentoring outcomes. Demographic attributes such as age, gender, occupation, and educational level can influence communication style, comfort, and alignment of expectations, whereas variables like religion, ethnicity, marital status, and income are often unrelated to educational mentoring goals and can introduce ethical concerns if used for matching. Personality trait theory suggests that underlying traits are relatively stable through adulthood, but matching based on self-reported traits may not accurately reflect mentoring behaviour, and similar trait profiles do not guarantee effective coaching. Consequently, in this work we focus on concrete, observable features such as domain experience, research skills, communication readiness, and mentoring goals, and use demographic markers only where they are clearly justified by programme objectives.

Taken together, two-sided matching methods provide rank-based pairwise allocations, clustering-based mentor systems offer similarity-driven assignments without full constraint modelling, and constrained clustering techniques illustrate how side information can be encoded in clustering objectives. The present approach builds on these strands by combining embedding-based similarity with constraint-aware group formation, treating demographic factors as configurable and programme-dependent rather than as fixed fairness targets, and explicitly modelling mentees’ research readiness and collaboration skills through targeted questionnaires.

3 Methodology

Our goal is to construct research-oriented mentor-mentee groups that are (i) feasible in terms of mentor capacity and scheduling, (ii) well aligned with the intended research focus of each group, and (iii) internally diverse with respect to mentees’ research readiness and study skills, while maintaining coherence in their long- and short-term goals. To achieve this, we model group formation as a constrained selection problem and apply a three-stage pipeline: (1) feasibility filtering based on experience and availability, (2) semantic retrieval of research-aligned candidates using text embeddings, and (3) diversity-aware refinement and final ranking based on questionnaire responses and goal-related embeddings. This structure mirrors modern matching approaches which separate hard constraints from preference modelling and optimization [3, 5].

3.1 Participant representation

Each participant is represented by a combination of structured attributes and vector embeddings derived from their questionnaire responses and profile information.

Structured attributes. For mentors, we consider their experience level, maximum mentee capacity, and weekly availability over discrete day-time blocks. For mentees, we use their self-reported experience level and availability in the same temporal grid. In addition, mentees complete a short research-readiness questionnaire covering five dimensions: confidence in conducting literature reviews, communication abilities, proficiency with research tools, deadline management, and domain knowledge. These five questionnaire scores form a low-dimensional feature space that we later use to promote diversity within groups.

Textual profiles and embeddings. To capture higher-level research preferences and developmental goals, we organize textual fields into two conceptual bundles. The first bundle describes a mentee’s research domain (e.g., skills, field of study, and research goals). The second bundle encodes their mentoring goals (short-term and long-term goals and expectations from a mentor). Each bundle is mapped into a fixed-length embedding vector using a pre-trained sentence embedding model, yielding a “domain” embedding and a “combined profile” embedding for every mentee. For each mentor-led group, we also embed the textual group description (e.g., project topic and mentoring focus) into the same vector space. This representation allows us to measure semantic similarity between groups and mentees in a continuous space, rather than relying only on categorical matches or ordinal preferences, as in classical two-sided matching [5, 6].

3.2 Similarity modelling

We distinguish between three types of compatibility:

Scheduling feasibility. First, we operationalize availability as a set of time intervals in each participant’s local time zone. Mentor and mentee schedules are transformed into a common reference frame, and a pair is considered feasible if they share at least one overlapping interval of at least one hour. This constraint is treated as strictly binary: pairs without sufficient overlap are never considered further.

Research alignment. Second, we measure research alignment between a mentee and a group by cosine similarity between the group’s description embedding and the mentee’s domain embedding. Intuitively, this score is high when the mentee’s skills, field, and stated research goals are semantically close to the group’s research theme. Cosine similarity in embedding space has been widely used in clustering-based mentor matching and document coclustering [5, 6].

Questionnaire distance and goal coherence. Third, to encourage heterogeneity in research readiness and study skills, we define a Euclidean distance between mentees in the five-dimensional questionnaire space. Large distances indicate complementary strengths and weaknesses. In contrast, coherence in goals and expectations is captured via the combined profile embeddings: mentees whose goal embeddings are close to each other and to the group’s mentoring focus are considered more compatible at the final ranking stage.

3.3 Group formation strategy

For a given mentor-led group with target size n_{matches} , we construct the mentee set S_g in three stages.

Feasibility filtering. We first select all mentees whose experience level matches the group’s required level and whose availability intersects with the mentor’s availability for at least one hour. This step enforces the fundamental practical constraints of experience and scheduling and removes clearly infeasible pairs before any semantic comparison.

Semantic retrieval of research-aligned candidates. Among the feasible mentees, we compute the cosine distance between each mentee’s domain embedding and the group’s description embedding. We then retain the top $4n_{\text{matches}}$ mentees with the smallest distance. This stage acts as a high-recall semantic filter that concentrates subsequent computation on mentees whose research interests and skills are well aligned with the group topic, similar in spirit to embedding-based retrieval used in prior clustering and matching approaches [5, 6].

Diversity-aware refinement and final ranking. From this semantically aligned pool, we build a refined candidate set of size $2n_{\text{matches}}$ using a greedy max–min strategy on the questionnaire distance. Starting from the most domain-aligned mentee, we iteratively add the mentee that maximizes the minimum Euclidean distance to all previously selected mentees in the questionnaire space. This heuristic spreads mentees along the axes of research confidence, communication, tool skills, deadline management, and domain knowledge, yielding a set of candidates that are well aligned with the group topic but not homogeneous in their study skills.

Finally, we rank this diversified candidate set using similarity in the combined profile embedding space, which encodes mentees’ short-term and long-term goals and expectations from a mentor. The top n_{matches} mentees under this ranking are recommended as the final group for the mentor. Conceptually, this last step ensures that, among research-aligned and questionnaire-diverse candidates, we prioritize those whose developmental goals are mutually compatible and well matched to the stated mentoring focus of the group.

3.4 Conceptual objective

Although we implement group formation via staged heuristics, the methodology is guided by an implicit multi-objective criterion. For each group g with mentee set S_g , we aim to:

- satisfy hard feasibility constraints on experience level and availability;
- maximize average research alignment between the group description and mentee domain embeddings;
- maintain coherence in mentees’ goals and expectations relative to the mentoring context.

This combination of semantic similarity, structured constraints, and diversity in research readiness distinguishes our approach from purely rank-based two-sided matching, which typically ignores rich profile information, and from clustering-only approaches that do not explicitly enforce scheduling or capacity constraints [3, 6].

3.5 Evaluation design

To assess the quality of the mentor-mentee groups produced by our matching framework, we utilize an online feedback questionnaire administered after the mentoring activity. The evaluation focuses on whether groups are (i) compatible in terms of interaction dynamics, (ii) responsive in coordinating research work, and (iii) able to maintain high-quality relationships over time.

3.5.1 Post-matching feedback instrument

After each mentoring cycle, mentees are invited to complete an online feedback form. The form is structured into three sections:

- **Compatibility**, capturing perceived ease of communication, trust in group members, and constructiveness of disagreements.
- **Responsiveness**, capturing timeliness of replies, adherence to deadlines, and availability of support when needed.
- **Relationship quality**, capturing overall openness, mutual trust, and respectful conflict resolution.

Each statement in the form is rated on a 5-point scale, where 1 indicates strong disagreement and 5 indicates strong agreement. For example, items such as “I felt confident relying on my group members and trusted them with their portions of the research” and “Members completed their assigned tasks on time and did not cause unnecessary delays to the research schedule” are used to quantify trust and reliability within the group.

3.5.2 Derived evaluation metrics

From the questionnaire responses we compute, for each group g :

- the *compatibility score*, defined as the mean of all compatibility items across mentees in g ;
- the *responsiveness score*, defined as the mean of all responsiveness items across mentees in g ;
- the *relationship quality score*, defined as the mean of all relationship items across mentees in g ;
- the *overall satisfaction score*, defined as the mean of all items across the three sections.

References

- [1] Mentor connect: Ai-driven career mentorship platform for personalized growth and skill development. *TIJER - International Research Journal*, 12(5):462–468, May 2025.
- [2] Hernie B. Deduro, Rosemarie T. Gepolani, Lance Winslet Banday, Mary Ann G. Erediano, and Alabena M. Kalim. Development of a web-based peer mentoring and support platform: Enhancing academic performance of struggling second-year education students through guided assistance and collaborative learning. *International Journal of Science and Research Archive*, 18(3):1233–1249, 2026.
- [3] Christian Haas and Margaret Hall. Two-sided matching for mentor-mentee allocations: Algorithms and manipulation strategies, March 2019.
- [4] Masayuki Okabe and Seiji Yamada. Clustering using boosted constrained k-means algorithm. *Frontiers in Robotics and AI*, Volume 5 - 2018, 2018.
- [5] Sweni Shah, Hamid S. Timorabadi, Sanjana Dasadia, Samreen Khatib Syed, and Doaa Muhammad. Board 47: A mentor-mentee matching algorithm to automate process of finding an ideal mentor for students. Proceedings of the 2024 ASEE Annual Conference & Exposition, Portland, Oregon, June 2024.
- [6] Mitesh Sharma, Amit Mishra, and Devean Purohit. A literature survey on coclustering and constraint clustering for text documents. *International Journal of Engineering Research & Technology (IJERT)*, 2(03), 2014. ETRASCT – 2014.
- [7] Nayan Verma, Pranav Vishwakarma, and Aradhana Negi. A study on the efficacy of the mentorship platform in education and career advancement. *International Journal of Research Publication and Reviews*, pages 5063–5067, 06 2024.